Does the counterterm α_2 double count the Fingers-of-God dispersions?

The redshift-space counterterm against the sampled velocity dispersions σ_v, template by template and in the posterior

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1 Summary

The question. The forecast's one-loop model carries two descriptions of nonlinear velocities along the line of sight. One is the Fingers-of-God blur: each bin's redshift distribution is convolved with a Gaussian whose width is set by a velocity dispersion σ_v, one per dispersion group, sampled under a Gaussian prior. The other is the line-of-sight counterterm α_2 of the DESI 2024 full-shape convention [1],

$$P_{\rm CT}(k,\mu) \;=\; \left(b_1 + f\mu^2\right) f\,\alpha_2\,\mu^2 k^2\,P_{\rm lin}(k), \tag{1}$$

with k the comoving wavenumber, μ the cosine of the angle to the line of sight, b_1 the bin's linear galaxy bias, $f = d\ln D/d\ln a$ the linear growth rate (D the linear growth factor, a the scale factor), $P_{\rm lin}$ the linear matter power spectrum, and α_2 in $h^{-2}\,{\rm Mpc}^2$, one value per group plus a free tilt in redshift in model B + tilts. The leading effect of the blur on the power spectrum is a damping $-\sigma_\chi^2 k_\parallel^2$ with $k_\parallel = k\mu$ and $\sigma_\chi = (1+z)\,\sigma_v/H(z)$ the dispersion in comoving distance: the same $k_\parallel^2$ shape as Eq. (1) (Section 2). So the question is whether the forecast pays twice for one effect: is α_2 degenerate with the σ_v already in the model?

The answer.

1. **At the template level, yes.** On every layout and at every cut, combinations of the seven σ_v templates reproduce 95.8–100% of each α_2 group template's information in the C^{-1} metric (C the band-power covariance), with medians 99.3–99.9%; the two templates are mirror images, so a positive α_2 looks like a smaller σ_v (Figure 2). Only the α_2 tilt columns are substantially new.

2. **In the posterior, the σ_v priors break the degeneracy at small cuts but not at large ones.** In the paper model (model B + tilts on the pessimistic linear-bias prior) the largest correlation between an α_2 parameter and a σ_v grows, on the thin layouts, from $|\rho| = 0.39$–0.46 at $k_{\rm max} = 0.075\,h/{\rm Mpc}$ to 0.81–0.88 at 0.2 (Figure 1).

3. **What α_2 costs.** Fixing α_2 and its tilts at zero raises ${\rm FoM}(w_0, w_a)$ of the paper model by a factor of 2.4–2.7 at 0.075 and 1.16–1.25 at 0.2 on the thin layouts; ${\rm FoM}(\Omega_m, 10^9 A_s)$ moves the same way. The gain comes entirely from the σ_v priors: with flat σ_v priors, fixing α_2 returns 0.83–1.01 of the paper model's FoM at every cut from 0.075 on.

4. **What dropping α_2 risks.** If the truth holds a DESI-prior-sized $\alpha_2 = 12.5\,h^{-2}\,\mathrm{Mpc}^2$ in every group, a fit without α_2 (model A + tilts) is shifted in (w_0, w_a), and σ_v on its priors absorbs almost none of it: the shift falls by 4% and 17% at 0.075 and by 35% and 41% at 0.2 on the two thin layouts. With flat σ_v priors those shifts fall by 88–97%. The wide layouts barely react either way.

5. **A wider σ_v prior trades the gain for the absorption.** Multiplying all seven σ_v prior widths by a common factor s, the fit without α_2 absorbs the $12.5\,h^{-2}\,\mathrm{Mpc}^2$ truth (every shift below $\sigma/3$) at $s = 1.84$ and 1 at $k_{\max} = 0.075\,h/\mathrm{Mpc}$ on the thin layouts, where it still keeps 2.1 and 2.8 times the paper model's FoM; at 0.1–0.2 it absorbs only at $s = 3.6$–6.0, where the FoM ratio is 0.96–1.22 and the gain is gone (Figure 3). Widening one sample's dispersions alone never suffices at 0.1 and above.

6. **Sampling σ_v can outgrow the pair list.** The data vector's cross pairs are fixed by an overlap rule at the fiducial σ_v ($O_{ij} \geq 10^{-6}$, Section 6). Chains over the current priors stay inside the list: the nearest unlisted pair reaches 7.6×10^{-7} at the $+2\sigma$ edge on $\Delta d_{\mathrm{com}} = 62$ Mpc, which crosses the threshold at $+2.1$ prior widths. With the widened priors of item 5 the crossing sits at 0.5–0.8 of the widened width, and with flat σ_v a DESI-size negative α_2 pushes $\Delta z = 0.02$ across at $|\alpha_2| \approx 16$–$19\,h^{-2}\,\mathrm{Mpc}^2$.

Here w_0 and w_a parameterize the dark-energy equation of state $w(a) = w_0 + w_a(1-a)$; Ω_m is the matter density today in units of the critical density; A_s is the amplitude of the primordial curvature power spectrum, sampled as $10^9 A_s$; and $\mathrm{FoM}(w_0, w_a) = [\det \mathrm{Cov}(w_0, w_a)]^{-1/2}$ is the inverse area of the marginalized (w_0, w_a) ellipse, likewise $\mathrm{FoM}(\Omega_m, 10^9 A_s)$. In short: one $k_\parallel^2$ effect, counted twice at the template level; the posterior keeps the two apart only as long as the σ_v priors do, and the whole dark-energy gain of dropping α_2 is exactly as strong as those priors. Whether to keep both, and on what basis for the σ_v priors, is an open decision (appendix "Open issues").

The setup. The data are the DESI angular clustering band powers C_ℓ^{gg} alone, computed without the Limber approximation, on four bin layouts: $\Delta d_{\mathrm{com}} = 62$ Mpc (equal comoving width) and $\Delta z = 0.02$, 0.05 and 0.2 (fixed redshift width); the first two are the thin layouts. The scan runs over $k_{\max} = 0.05$ to $0.2\,h/\mathrm{Mpc}$, each the $z = 0$ value of a per-bin cut at $k_{\max}/D(\bar{z}_i)$, with $\bar{z}_i$ the mean redshift of bin i. Every number and figure of this note comes from the final section of `EXAMPLE_EVALUATE_FISHER_MODEL_B.ipynb`; the per-cut values are in the two figure folders' `data.json`.

2 Why the question is natural

Redshift is blurred distance. The forecast places each galaxy at its measured redshift. Inside groups and clusters, galaxies orbit with random speeds of several hundred km/s, so a galaxy's measured redshift carries a Doppler shift on top of its distance: in a redshift map a cluster stretches into a finger pointing at the observer, the Fingers of God [3]. A line-of-sight velocity v at redshift z displaces the galaxy by

$$\delta\chi \;=\; \frac{(1+z)\,v}{H(z)}, \tag{2}$$

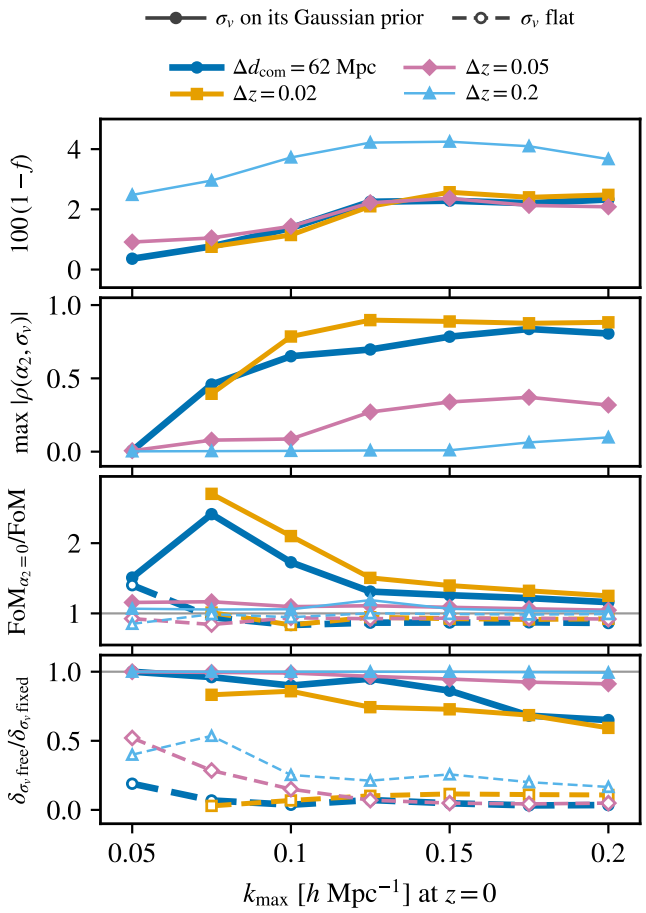


Figure 1: How far the counterterm α_2 and the Fingers-of-God dispersions σ_v describe the same band powers, against the scale cut, per layout (color, line width and marker; legend). Top panel: $100\,(1-f)$, the share of an α_2 group template's information in the C^{-1} metric that no combination of the seven σ_v templates reproduces, for the group where that share is largest; this panel is built from the templates and the covariance alone and does not depend on any prior. Second panel: the largest $|\rho|$ between an α_2 parameter and a σ_v in the posterior of the paper model, model B + tilts on the pessimistic linear-bias prior. Third panel: $\mathrm{FoM}_{\alpha_2=0}/\mathrm{FoM}$, the $\mathrm{FoM}(w_0, w_a)$ with α_2 and its tilts fixed at zero over the paper model's, with σ_v on its Gaussian priors (solid, filled markers) or flat (dashed, open markers). Bottom panel: $\delta_{\sigma_v\,\mathrm{free}}/\delta_{\sigma_v\,\mathrm{fixed}}$, with δ the (w_0, w_a) shift of model A + tilts, which has no α_2, fitted to a truth with $\alpha_2 = 12.5\,h^{-2}\,\mathrm{Mpc}^2$ in every group: the shift with σ_v free over the shift with σ_v fixed at its fiducial, with the same two line styles. Gray: one. $\Delta z = 0.02$ keeps no band at $0.05\,h/\mathrm{Mpc}$. $k_{\rm max}$ quoted at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$.

with χ the comoving distance and $H(z)$ the expansion rate. A dispersion σ_v of these velocities blurs distances by $\sigma_\chi = (1+z)\,\sigma_v/H(z)$: for the groups here, 5–6 Mpc, about $4\,h^{-1}$Mpc (`notes/cross_correlations`, which derives the convolution and its sizes).

Blurring the window equals blurring the field. The likelihood implements the blur by convolving each bin's redshift distribution $n_i(z)$ with a Gaussian of width $(1+\bar{z}_i)\,\sigma_v/c$ in redshift [5]. That window-side convolution is the same operation as damping the three-dimensional field, as long as the velocities are a random scatter uncorrelated with the density. Write the observed galaxy field as the true one with each galaxy moved by $\delta\chi$ drawn from a distribution p: along one line of sight, $\delta_g^{\rm obs}(\chi_{\rm obs}) = \int d\chi\, p(\chi_{\rm obs}-\chi)\,\delta_g(\chi)$. A bin's map integrates the observed field over its radial window W_i:

$$\begin{aligned}\int d\chi_{\rm obs}\, W_i(\chi_{\rm obs})\,\delta_g^{\rm obs}(\chi_{\rm obs}) \;&=\; \int d\chi \Big[\int d\chi_{\rm obs}\, W_i(\chi_{\rm obs})\, p(\chi_{\rm obs}-\chi)\Big]\,\delta_g(\chi)\\ &=\; \int d\chi\,\big(W_i{*}p\big)(\chi)\,\delta_g(\chi)\,: \end{aligned}\tag{3}$$

swapping the order of integration moves the convolution from the field onto the window. Blurring every bin's $n(z)$ and keeping the true field is therefore the same map as keeping sharp bins and blurring the field, which is how the likelihood's σ_v reach the spectra.

The damping and its leading order. On the field side, a convolution along the line of sight multiplies each Fourier mode by the distribution's transform: for a Gaussian p of width σ_χ, the density picks up $\exp(-k_\parallel^2\sigma_\chi^2/2)$ and the power spectrum $\exp(-k_\parallel^2\sigma_\chi^2)$, with $k_\parallel = k\mu$ the line-of-sight wavenumber. Applied to the linear redshift-space (Kaiser [4]) power, to leading order in $k_\parallel^2\sigma_\chi^2$,

$$\big(b_1+f\mu^2\big)^2 P_{\rm lin}\, e^{-k_\parallel^2\sigma_\chi^2} \;\simeq\; \big(b_1+f\mu^2\big)^2 P_{\rm lin} \;-\; \sigma_\chi^2\, k_\parallel^2\,\big(b_1+f\mu^2\big)^2 P_{\rm lin}.\tag{4}$$

The counterterm, Eq. (1), is $f\,\alpha_2\,k_\parallel^2\,(b_1+f\mu^2)\,P_{\rm lin}$. Both are $k_\parallel^2$ times the linear power; their μ structures differ by exactly one factor of $(b_1+f\mu^2)$, and a band power averages much of that difference away. The signs are opposite where both amplitudes are positive: a positive α_2 raises the spectrum where a larger σ_v lowers it, so a positive α_2 looks like a *smaller* σ_v. The sizes are comparable too: $\sigma_\chi \approx 4\,h^{-1}$Mpc gives $\sigma_\chi^2 \approx 16\,h^{-2}\,{\rm Mpc}^2$, the order of the DESI prior width $12.5\,h^{-2}\,{\rm Mpc}^2$ of α_2 at $k_{\rm max} = 0.2\,h/{\rm Mpc}$.

Why both exist. In a three-dimensional full-shape analysis there is no radial window to convolve, so there is no sampled σ_v: the counterterm α_2 is that analysis's effective description of small-scale velocities, Fingers of God included [1, 2]. This forecast works on projected spectra, where the window-side convolution is natural and σ_v already carries the effect; α_2 was added with the rest of the DESI counterterm set. A setup with α_0 only – the density counterterm, which multiplies $k^2 P_{\rm mm}$ with no μ – is the natural alternative, and this note measures what each choice costs.

The seven dispersions and their priors. Each bin's σ_v is its dispersion group's value (Table 1): the measured dispersion curve of the source forecast's simulated galaxies steps

inside BGS, so BGS takes two values, split at $\bar{z} = 0.255$, giving seven dispersion groups for six bias groups. Each prior is a Gaussian centred on the measured value, five times as wide as that measurement's per-bin error bar, and truncated at zero; the defaults and priors live in `likelihood/params_desi*.yaml` (the `DESI_SIGV` parameters, one sampled value per group), and their provenance is the section "Where the velocity dispersions come from" of `notes/cross_correlations`. The same seven values and widths serve every layout.

dispersion group	σ_v [km/s]	prior width [km/s]
BGS, $\bar{z}_i < 0.255$	371	145
BGS, $\bar{z}_i \geq 0.255$	358	105
LRG1	357	70
LRG2	348	50
LRG3	343	50
ELG1	304	40
ELG2	289	30

Table 1: The fiducial velocity dispersion of each group and the width $\Delta\sigma_{v,g}$ of its Gaussian prior.

3 Templates, and how to measure redundancy

A parameter's template. A parameter p's template is the derivative of the data vector, $\partial C_\ell/\partial p$ over every kept band of every spectrum. To first order it is the fingerprint p leaves on the data: the fit can distinguish two parameters only as far as their templates differ. Each parameter is measured in units of its prior width Δ_p (written Δ, never σ, since σ_v is itself a dispersion): here $\Delta\alpha_2 = 0.5/k_{\rm max}^2$, the model B prior width, quoted at $k_{\rm max} = 0.2\,h/{\rm Mpc}$ where it is $12.5\,h^{-2}\,{\rm Mpc}^2$, and $\Delta\sigma_{v,g}$ each dispersion group's prior width (Table 1). Divided by each band's error $\sigma_\ell = \sqrt{\mathsf{C}_{ii}}$, the scaled template $\sigma_\ell^{-1}\,\partial C_\ell/\partial(p/\Delta_p)$ reads directly as how many error bars the parameter can move each band while staying within its prior; Figure 2 draws the templates this way, following Fig. 1 of Obied et al. [7]. The α_2 templates here are group templates: the response to the group's value, all bins of the group moving together (and, for the tilts, moving along a straight line in redshift).

The leftover, measured in the data's own error bars. Can an α_2 group template be rebuilt as a weighted sum of the seven σ_v templates? The leftover of the best weights c_g is

$$r_\ell \;=\; \frac{\partial C_\ell}{\partial(\alpha_2/\Delta\alpha_2)} \;-\; \sum_g c_g\,\frac{\partial C_\ell}{\partial(\sigma_{v,g}/\Delta\sigma_{v,g})}, \tag{5}$$

the sum running over the seven dispersion groups. It should not be measured band by band: a mismatch of $0.1\sigma_\ell$ on a precise band is visible where the same mismatch on a noisy band is not. The natural norm is the data's own, $r^{\mathsf{T}}\mathsf{C}^{-1}r$ with r the vector of the r_ℓ, the χ^2 the leftover would contribute, and the weights c_g minimize it. Writing a for the scaled α_2 template, the first term of Eq. (5), the redundancy fraction is the share of the template's information that the σ_v templates reproduce,

$$f \;=\; 1 \;-\; \frac{\min_c\, r^{\mathsf{T}}\,\mathsf{C}^{-1}\,r}{a^{\mathsf{T}}\,\mathsf{C}^{-1}\,a}. \tag{6}$$

$f = 1$ means the data cannot tell α_2 from that combination of dispersions at all. (The letter f follows the figure axis; the growth rate is written f only inside Eqs. (1) and (4).)

A three-band toy. Take three bands with unit error bars and two templates, $a = (1, 2, 3)$ and $s = (2, 4, 6)$. Then $a = s/2$ exactly: the leftover is zero and $f = 1$; whatever signal a could produce, $s/2$ produces identically, and the data cannot separate the two parameters. Now take $s = (2, 4, 5)$, which differs from $2a$ only on the third band. The best single weight is $c = 5/9$, the leftover is $r = (-1, -2, 2)/9$ with $r^{\mathsf{T}} r = 1/9$, and $f = 1 - (1/9)/14 = 0.992$. Whether that mismatch matters is decided by the third band's error bar. If the third band is noisy, the fit matches the first two bands almost exactly and hides the whole mismatch inside the third band's error: the leftover χ^2 falls toward zero and $f \to 1$. If the third band is precise, the fit is pinned there ($c \to 3/5$) and the mismatch is pushed onto the first two bands as a visible leftover. Either way, a mismatch is only as detectable as the error bars of the bands where it must appear – which is what the C^{-1} weighting in Eq. (6) counts.

One parameter against one: $f = \rho^2$. With one α_2 amplitude, one σ_v and nothing else free (templates a and s), the two-parameter Fisher matrix is built from $a^{\mathsf{T}}\mathsf{C}^{-1}a$, $a^{\mathsf{T}}\mathsf{C}^{-1}s$ and $s^{\mathsf{T}}\mathsf{C}^{-1}s$, and the correlation of its inverse is $\rho = -\,a^{\mathsf{T}}\mathsf{C}^{-1}s\,/\sqrt{(a^{\mathsf{T}}\mathsf{C}^{-1}a)(s^{\mathsf{T}}\mathsf{C}^{-1}s)}$, so $\rho^2 = f$. Mirror-image templates ($a^{\mathsf{T}}\mathsf{C}^{-1}s < 0$) give a *positive* posterior correlation: the direction the data cannot see has α_2 and σ_v rising together, each undoing the other's change of the spectra. The posterior correlation of the full model is a different number from f: it marginalizes every other parameter – cosmology, biases, the other counterterms – which can soak up part of either template, and it includes the priors, which hold parameters the data would let drift. That is why f can sit at 0.99 while $|\rho|$ ranges from 0.003 to 0.90 across cuts and layouts.

Many against many: canonical correlations. For the full sets – six α_2 group columns (plus tilts) against seven σ_v columns – the squared canonical correlations generalize ρ^2: they are the cosines of the principal angles between the two template subspaces in the C^{-1} metric, the per-direction version of Eq. (6).

4 Results

The templates are mirror images. Figure 2 draws the α_2 and σ_v templates of three $\Delta d_{\rm com} = 62$ Mpc spectra, each scaled by its prior width. The two templates track each other band by band with opposite signs, at a few tenths of σ_ℓ per band. The bottom row is the leftover r_ℓ of Eq. (5): after the best combination of the seven σ_v templates, fitted over the whole data vector, $|r_\ell|/\sigma_\ell$ stays below 0.04 per band in the BGS panel and below 0.013 in the LRG panels at the $0.2\,h$/Mpc cut, and below 0.004 everywhere at 0.075: templates of a few tenths of an error bar cancel to a few thousandths.

The redundancy against the cut. Over all four layouts and every cut of the scan, f of the six α_2 group columns lies between 0.958 and 1.000, with per-cut medians of 0.993–0.999; the squared canonical correlations of the six-column α_2 set with the seven-column σ_v set lie in the same range, 0.958–1.000. The top panel of Figure 1 draws $100\,(1 - f)$ for the least redundant group: even there, at most 4.3% of the template's information is new. The α_2 *tilt* columns

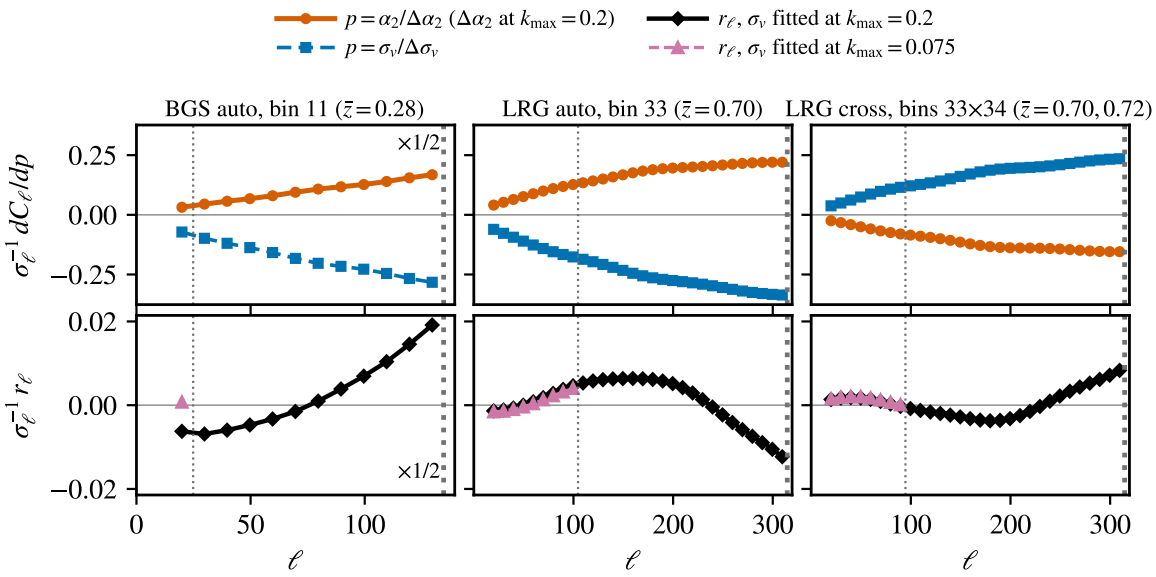


Figure 2: The α_2 and σ_v templates of three $\Delta d_{\rm com} = 62$ Mpc spectra – a BGS auto spectrum ($\bar{z} = 0.28$), an LRG auto spectrum ($\bar{z} = 0.70$) and that bin's cross spectrum with the next LRG bin – in units of each band's error $\sigma_\ell = \sqrt{\mathsf{C}_{ii}}$, on the bands the shipped mask keeps; the BGS column is divided by the round factor written in its panels, and each row shares one y axis. Top row: the scaled templates $\sigma_\ell^{-1}\, \partial C_\ell/\partial p$, each parameter in units of its prior width: $p = \alpha_2/\Delta\alpha_2$ of the bin's group, with $\Delta\alpha_2$ the prior width at $k_{\max} = 0.2\, h/\mathrm{Mpc}$ (vermillion solid, circles), and $p = \sigma_v/\Delta\sigma_v$ of the bin's dispersion (blue dashed, squares). Bottom row: $\sigma_\ell^{-1} r_\ell$, the leftover of the α_2 template after its best combination of the seven σ_v templates, Eq. (5), fitted over the whole data vector in the C^{-1} metric on the bands a cut keeps and drawn on those bands, for $k_{\max} = 0.2$ (black, diamonds) and $0.075\, h/\mathrm{Mpc}$ (pink dashed, triangles). Dotted vertical lines: where the shown spectrum's kept bands end at 0.075 (thin) and at 0.2 (thick). $k_{\max}$ quoted at $z = 0$; each bin is cut at $k_{\max}/D(\bar{z}_i)$.

are much less redundant – their per-cut medians lie between 0.03 and 0.24 at every cut except the smallest cut on $\Delta d_{\rm com} = 62$ Mpc, where only three tilt columns keep any band – because a straight line in redshift across a group is a shape no combination of seven group-constant dispersions reproduces. At the template level, then, the α_2 group values duplicate information the σ_v already carry; the tilts are the genuinely new freedom.

The posterior of the paper model. The second panel of Figure 1 shows the largest $|\rho(\alpha_2, \sigma_v)|$ in the paper model's posterior. On the thin layouts it grows from 0.46 ($\Delta d_{\rm com} =$ 62 Mpc) and 0.39 ($\Delta z = 0.02$) at $k_{\max} = 0.075\, h/\mathrm{Mpc}$ to 0.81 and 0.88 at 0.2, peaking at 0.84 and 0.90 at intermediate cuts; on $\Delta z = 0.05$ it stays at 0.08–0.37, and on $\Delta z = 0.2$ below 0.10. Wherever the correlation is appreciable, the pair is LRG3's α_2 with LRG3's own dispersion, and its sign is positive, as mirror-image templates give (Section 3). The correlations stay below one because the σ_v priors hold their end of the degenerate direction: with α_2 free, every σ_v posterior keeps 0.60–1.00 of its prior width, so the dispersions stay prior-dominated while α_2

takes the $k_\parallel^2$ information. With α_2 and its tilts fixed, the data begin to measure the dispersions: on the thin layouts the best-measured σ_v tightens to 0.25–0.74 of its prior width at cuts of 0.075 and above.

What α_2 costs. The third panel of Figure 1 draws $\mathrm{FoM}_{\alpha_2=0}/\mathrm{FoM}$: the paper model with α_2 and its tilts fixed at zero, over the paper model. On the thin layouts the fixed fit has 2.41 and 2.70 times the $\mathrm{FoM}(w_0, w_a)$ at $k_{\max} = 0.075\,h/\mathrm{Mpc}$, 1.73 and 2.10 at 0.1, and 1.16 and 1.25 at 0.2; $\mathrm{FoM}(\Omega_m, 10^9 A_s)$ gains similar factors (2.27 and 2.19 at 0.075, 1.18 and 1.25 at 0.2). On $\Delta z = 0.05$ the gain is 1.05–1.17, and on $\Delta z = 0.2$ at most 1.07 at every cut except 0.125, where it reaches 1.19. The dashed curves isolate where the gain comes from: with the σ_v priors removed (flat), fixing α_2 returns 0.83–1.01 of the paper model's FoM at every cut from 0.075 on – no gain at all. Fixing α_2 therefore does not add information of its own; it hands the $k_\parallel^2$ direction to the dispersions, and the constraint the fixed fit gains is exactly the constraint the σ_v priors supply.

What dropping α_2 risks. The absorption test fits model A + tilts, which has no α_2, to a truth with $\alpha_2 = 12.5\,h^{-2}\,\mathrm{Mpc}^2$ in every group – the DESI prior width at $k_{\max} = 0.2\,h/\mathrm{Mpc}$. If σ_v could stand in for α_2 in practice, freeing it would absorb the shift. It barely does: with σ_v on its Gaussian priors, the (w_0, w_a) shift falls from the fixed-σ_v value by 4% at 0.075 ($0.335 \to 0.322$ ellipse units) and 35% at 0.2 ($1.65 \to 1.07$) on $\Delta d_{\mathrm{com}} = 62$ Mpc, and by 17% ($0.305 \to 0.254$) and 41% ($2.60 \to 1.54$) on $\Delta z = 0.02$; on $\Delta z = 0.05$ by at most 9%, and on $\Delta z = 0.2$ the shift never exceeds 0.016σ to begin with. With flat σ_v priors the same shifts, at cuts of 0.075 and above, fall by 88–97% (bottom panel of Figure 1): the dispersions *can* mimic the counterterm, but only by moving far outside their priors. Fitting the $\alpha_2 = 12.5\,h^{-2}\,\mathrm{Mpc}^2$ truth with the σ_v templates alone makes this concrete: on $\Delta d_{\mathrm{com}} = 62$ Mpc at 0.2 the truth carries $\Delta\chi^2 = 162$, of which free dispersions leave 1.6 – but the mimicking shift is -2.5 prior widths (-75 km/s, always the ELG2 dispersion, whose prior is tightest), and with the Gaussian priors the total left, data residual plus prior penalty, is 14. Across the thin layouts at cuts of 0.075 and above, the mimicking shift is 1.9–2.6 prior widths (57–78 km/s). The shifts scale linearly with the truth: a truth at one model B prior width ($0.5/k_{\max}^2$, $88.9\,h^{-2}\,\mathrm{Mpc}^2$ at 0.075) gives 7.1 times the $12.5\,h^{-2}\,\mathrm{Mpc}^2$ shifts there.

5 How wide must the σ_v priors be?

The knob. The absorption test leaves a natural follow-up question: if σ_v keeps a Gaussian prior, how wide must that prior be for the dispersions to absorb a DESI-size α_2? The two preceding results frame the answer before any scan. Mimicking $\alpha_2 = 12.5\,h^{-2}\,\mathrm{Mpc}^2$ costs the dispersions 1.9–2.6 prior widths, so widening the priors by a factor of a few makes that move cheap and the shift disappears. But the FoM gain of dropping α_2 comes entirely from those same priors, so the same widening surrenders the gain. A single knob – a common factor s on all seven prior widths of Table 1 – therefore slides the fit between two ends: at $s = 1$ the full gain and almost no absorption, with flat priors the full absorption and no gain. The scan locates where between the two ends each layout and cut sits.

The scan. The fit is the one of the absorption test – model A + tilts on the pessimistic linear-bias prior, which carries α_0 but no α_2: the α_0-only setup of Section 2 – fitted to the truth $\alpha_2 = 12.5\,h^{-2}\,\mathrm{Mpc}^2$ in every group, with all seven σ_v prior widths multiplied by $s \in \{1, 1.5, 2, 3, 5, 10\}$ and flat. Each s is scored by $\delta_{\max}$, the largest of the (w_0, w_a) shift in its ellipse and the Ω_m and A_s shifts in their own σ, and by $\mathrm{FoM}_{\alpha_0\,\mathrm{only}}/\mathrm{FoM}$, that fit's $\mathrm{FoM}(w_0, w_a)$ over the paper model's at $s = 1$. The passing s is the smallest s with every shift below $\sigma/3$, interpolated linearly in $\log s$ between the scanned values. Figure 3 draws the scan; every cut, the per-shift values and the one-sample variants below are in that folder's `data.json`.

At $k_{\max} = 0.075\,h/\mathrm{Mpc}$ the default priors almost suffice. $\Delta d_{\mathrm{com}} = 62$ Mpc passes at $s = 1.84$, keeping 2.13 times the paper model's FoM; $\Delta z = 0.02$ passes at $s = 1$, keeping 2.77. $\Delta z = 0.05$ passes at $s = 1$ at every cut up to 0.15 (FoM ratios 1.12–1.23) and needs $s \approx 2.9$ at 0.175–0.2, where the ratio is 1.02–1.03; $\Delta z = 0.2$ passes at $s = 1$ everywhere (1.05–1.25).

At 0.1–0.2 on the thin layouts the two options converge. There the passing factor is $s = 3.56$–3.83 on $\Delta d_{\mathrm{com}} = 62$ Mpc and 4.83–5.99 on $\Delta z = 0.02$: ELG2's 30 km/s prior becomes about 110–180 km/s, a third to more than half of its fiducial dispersion. At those widths the FoM ratio is 0.96–1.22: the widened α_0-only fit and the paper model constrain dark energy essentially equally, and the gain that motivated dropping α_2 is gone.

One sample's dispersions alone do not absorb it. The truth holds α_2 in every group, so the mimicking must too. Widening only the three LRG dispersions fails wherever $s = 1$ does not already pass: on $\Delta d_{\mathrm{com}} = 62$ Mpc at 0.2, $\delta_{\max}$ falls only from 1.16 to 1.12 even with the LRG priors flat. Widening only the two ELG dispersions passes on $\Delta d_{\mathrm{com}} = 62$ Mpc at 0.075 ($s = 2.94$, ratio 2.24) and on $\Delta z = 0.05$ at 0.175–0.2, but fails on the thin layouts at 0.1 and above. Absorbing α_2 needs all the dispersions widened together.

6 The pair list at the edges of the σ_v prior

The rule. The data vector measures a fixed list of cross spectra, chosen by the Fingers-of-God overlap rule (`scripts/make_desi_pairs.py`; `notes/cross_correlations` derives it). With the normalized overlap of two bins' blurred $n(z)$,

$$O_{ij} \;=\; \frac{\int dz\, n_i(z)\, n_j(z)}{\sqrt{\int dz\, n_i^2(z)\, \int dz\, n_j^2(z)}}, \tag{7}$$

1 for two identical shapes, 0 for disjoint ones and, at high ℓ, close to the correlation coefficient of the two bins' spectra, the list holds DESI's measured pairs plus every pair with $O_{ij} \geq 10^{-6}$ at the fiducial σ_v. The list and the covariance are built once, at the fiducial dispersions; a sampled σ_v changes every kernel but never the list.

Raising σ_v pushes unlisted pairs toward the threshold. A larger σ_v widens every kernel, so every overlap grows – the listed ones, and those of the pairs outside the list, bins of one sample two apart, whose blurred kernels are separated by a full bin. Figure 4 follows the largest unlisted overlap as every dispersion group moves to $\sigma_v^{\mathrm{fid}} + k\,s\,\Delta\sigma_v$, with k the offset in

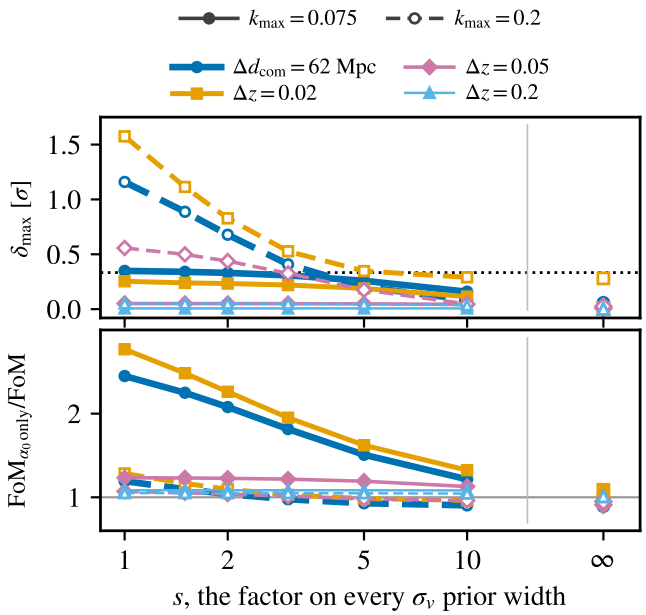


Figure 3: The σ_v prior-width scan: the α_0-only fit (model A + tilts on the pessimistic linear-bias prior) fitted to a truth with $\alpha_2 = 12.5\,h^{-2}\,\mathrm{Mpc}^2$ in every group, against s, the common factor on all seven σ_v prior widths (log axis; the flat-σ_v points sit at the right edge, past the gap), per layout (color, line width and marker; legend) at $k_{\max} = 0.075\,h/\mathrm{Mpc}$ (solid, filled markers) and 0.2 (dashed, open markers). Top panel: $\delta_{\max}$, the largest of the (w_0, w_a) shift in its ellipse and the Ω_m and A_s shifts in their own σ; dotted: $\sigma/3$. Bottom panel: $\mathrm{FoM}_{\alpha_0\,\rm only}/\mathrm{FoM}$, that fit's $\mathrm{FoM}(w_0, w_a)$ over the paper model's, model B + tilts with α_2 free and σ_v at $s = 1$; gray: one. $k_{\max}$ quoted at $z = 0$; each bin is cut at $k_{\max}/D(\bar{z}_i)$.

prior widths and s the width factor of Section 5. At the fiducial it is zero on every layout: no unlisted kernels meet. At $k = 1$, 2 and 3 with $s = 1$ it reaches 1.7×10^{-10}, 7.6×10^{-7} and 2.9×10^{-5} on $\Delta d_{\rm com} = 62$ Mpc (a pair of BGS bins two apart), and 8.6×10^{-10}, 5.0×10^{-8} and 7.9×10^{-7} on $\Delta z = 0.02$ (ELG2, then LRG3 bins two apart), while $\Delta z = 0.05$ and 0.2 stay at zero. The threshold is crossed at $k = 2.06$ on $\Delta d_{\rm com} = 62$ Mpc, 3.11 on $\Delta z = 0.02$, 12.5 on $\Delta z = 0.05$ and nowhere on $\Delta z = 0.2$. With the priors widened by $s = 4$, the order Section 5 requires at 0.1–0.2, the thin layouts cross already at $k \approx 0.5$–0.8.

The paper model stays inside the list. At the $+2\sigma$ edge of the current priors the largest unlisted overlap is 7.6×10^{-7} on $\Delta d_{\rm com} = 62$ Mpc – a factor 1.3 below the threshold – and 5.0×10^{-8} on $\Delta z = 0.02$; at +2 posterior widths of the paper model it is the same (at 0.075, where the posterior equals the prior) or smaller (at 0.2). A chain over the current priors therefore keeps every pair it needs in the list; $\Delta d_{\rm com} = 62$ Mpc crosses only just beyond, at +2.1 prior widths.

What α_2 asks of σ_v. Only a *negative* α_2 drives the dispersions up: a positive α_2 looks like a smaller σ_v (Section 4), so its mirror image raises them. In the linear response of the

fit without α_2, one unit of α_2 moves the most responsive dispersion by 0.1–2.2 km/s with the priors on (the pull grows with the cut) and by 5.4–7.2 km/s with flat priors (on every layout but $\Delta z = 0.2$). The $|\alpha_2|$ at which an unlisted pair first reaches 10^{-6} is, with flat σ_v: 16–19 $h^{-2}\,\mathrm{Mpc}^2$ on $\Delta z = 0.02$ – 1.3–1.5 times the DESI prior width $12.5\,h^{-2}\,\mathrm{Mpc}^2$ – through a pair of ELG2 bins two apart; 53–59 on $\Delta d_{\mathrm{com}} = 62$ Mpc (ELG1); 105–133 on $\Delta z = 0.05$; and 394–462 on $\Delta z = 0.2$. With the current priors it is at least $68\,h^{-2}\,\mathrm{Mpc}^2$ at $k_{\mathrm{max}} = 0.2\,h/\mathrm{Mpc}$ and at least $1000\,h^{-2}\,\mathrm{Mpc}^2$ at 0.075.

The caution for chains. The pair list and the covariance are built once, at the fiducial σ_v. The range of α_2 a chain treats as reasonable therefore bounds how far σ_v travels when it has to absorb α_2, and the widened or flat σ_v of Section 5 is exactly the regime where the list stops being complete: dispersions high enough to mimic a DESI-size negative α_2 also raise unlisted pairs – bins two apart – past the threshold, so more cross-correlations would have to enter the data vector. Either the chain bounds σ_v (or α_2) within the ranges above, or the list and covariance are rebuilt at the prior edge.

The listed pairs' own overlaps. The complementary test – keeping only the listed pairs whose overlap reaches a floor O_{min}, DESI-measured pairs included – drops, for every floor up to 10^{-3}, one pair: the BGS–LRG1 boundary pair of $\Delta d_{\mathrm{com}} = 62$ Mpc ($O = 3.4 \times 10^{-7}$), at a cost of 0.1% of FoM(w_0, w_a) at $k_{\mathrm{max}} = 0.2\,h/\mathrm{Mpc}$ and nothing elsewhere, so none of this note's conclusions rests on a low-overlap pair. The study and its limits are in `notes/cross_correlations`, section "How much do the low-overlap pairs carry?".

7 What this means

At the template level, α_2 and the sampled σ_v carry the same $k_\parallel^2$ information: in that sense the model double counts one physical effect, and the α_2 group values add almost no template direction the dispersions do not already span. Only the α_2 tilts are new.

In the posterior, the Gaussian σ_v priors break the degeneracy. They are tight enough (10–40% of the fiducial dispersions) that the dispersions cannot wander along the degenerate direction, so the joint fit stays well conditioned at every cut, with the largest α_2–σ_v correlation reaching 0.9 only at the largest cuts on the thin layouts.

Keeping both parameters has a definite division of labor: α_2, with its wide prior, takes the $k_\parallel^2$ information, and the dispersions fall back to their priors (0.60–1.00 of the prior width with α_2 free). The data then constrain nonlinear velocities only as far as the α_2 prior allows, and the σ_v priors contribute almost nothing to the FoM.

Dropping α_2 gains a factor of 2.4–2.7 in FoM(w_0, w_a) at $k_{\mathrm{max}} = 0.075\,h/\mathrm{Mpc}$ and 1.16–1.25 at 0.2 on the thin layouts – a gain that is exactly as good as the σ_v priors, since it vanishes when those priors are removed. The same move carries the bias risk of the absorption test: if a DESI-prior-sized α_2 is present in the truth, a fit without α_2 is shifted by up to 0.34 ellipse units at 0.075 and 1.6–2.6 at 0.2 on the thin layouts, and the dispersions on their priors absorb at most 4–41% of it. The wide layouts barely care either way: their gains are at most $\sim$20% and their shifts at most 0.43σ ($\Delta z = 0.05$ at 0.2).

These are the trade-offs as measured; whether the forecast keeps α_2, drops it in favor of the dispersions, or moves to a setup with α_0 only is not decided in this note (appendix "Open

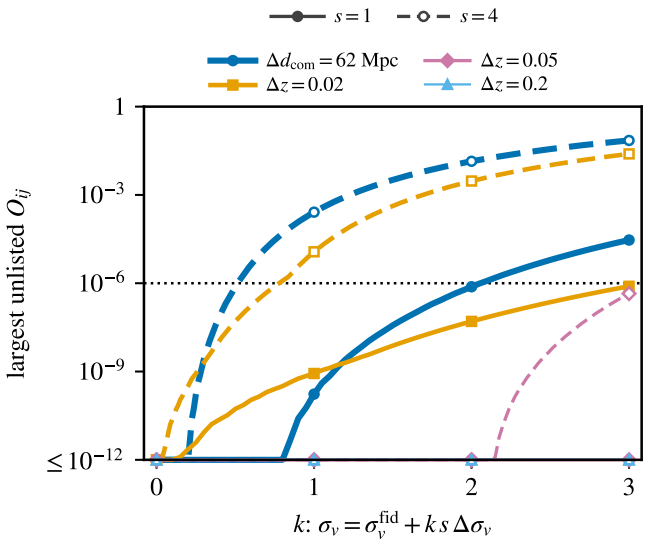


Figure 4: How far the pair rule holds away from the fiducial dispersions: the largest overlap O_{ij} of Eq. (7) over the pairs *outside* each layout's pair list, against the offset k of every σ_v group from its fiducial, in units of its prior width times the width factor s ($\sigma_v = \sigma_v^{\rm fid} + k\, s\, \Delta\sigma_v$), per layout (color, line width and marker; legend). Solid, filled markers: $s = 1$, the current priors; dashed, open: $s = 4$, the order of the widths that let σ_v absorb a DESI-size α_2; markers at $k = 0$, 1, 2, 3. Dotted: the rule's threshold $O_{ij} = 10^{-6}$, above which a pair belongs in the data vector. Overlaps below the axis floor, zero included (kernels that do not meet), are drawn on the floor.

issues").

A Method and checks

Machinery. Everything runs on the model B notebook's final section (Section 1 names the file): the final-tier derivative caches of that notebook (five-point stencils; the α_2 columns are exact one-difference templates, since the term is linear in its amplitude), the Gaussian band-power covariance restricted to the bands each cut keeps, and the notebook's Fisher systems $F = \mathsf{D}^{\mathsf{T}}\mathsf{C}^{-1}\mathsf{D} + F_{\rm priors}$, with D the derivative matrix and $F_{\rm priors}$ the Gaussian priors of the paper model (the pessimistic linear-bias prior, the one-loop offset and counterterm priors, and the σ_v priors of Table 1). Posterior correlations are read off F^{-1}; the fixed-α_2 and flat-σ_v variants remove the α_2 columns or the σ_v prior rows and rebuild F.

The absorption test is a linear response. The shift of model A + tilts under an α_2 truth is the Fisher linear response $\delta p = F^{-1}\mathsf{D}^{\mathsf{T}}\mathsf{C}^{-1}\,\delta m$, with δm the truth's change of the data vector, scored in the fit's marginalized (w_0, w_a) ellipse (σ for one parameter) – not a refit. Shifts of one to two ellipse units extrapolate that linear response accordingly.

Checks. The analysis reproduces, to the printed precision (0.0 relative difference), the paper model's FoM(w_0, w_a) and FoM($\Omega_m, 10^9 A_s$) of the model B notebook's `fom_kmax_models` figure data at every cut, and the model A + tilts α_2-truth shifts of the notebook's break-test scores. The σ_v-template fits of the $\alpha_2 = 12.5\,h^{-2}\,\mathrm{Mpc}^2$ truth satisfy the linear-algebra identities of Eq. (6) by construction; the notebook's own self-checks on the α_2 columns (the data vector returns bitwise after the template evaluation; forward and backward differences agree) are listed with that notebook.

B Open issues

- **Keep or drop α_2.** The trade is measured in this note – a factor 2.4–2.7 of FoM(w_0, w_a) at $k_{\max} = 0.075\,h/\mathrm{Mpc}$ on the thin layouts against the absorption shifts if a DESI-sized α_2 is real – but the decision is not made here. The prior-width scan sharpens what it is: in effect a choice between keeping α_2 and an α_0-only model whose σ_v prior widths are multiplied by the passing s of Section 5, and the two converge in FoM at $k_{\max} \geq 0.1$ on the thin layouts, so the cut at which the forecast is quoted matters to the choice.

- **The basis of the σ_v priors.** The gain from dropping α_2 is exactly as strong as the σ_v priors, so it inherits their basis: Gaussians five times as wide as the per-bin error bars of the source forecast's measurement on simulated galaxies, truncated at zero (`notes/cross_correlations`). How defensible that width is for the real samples, and whether a Gaussian blur is the right shape for the small-scale velocity distribution, are open.

- **No redshift-space one-loop terms.** The sampled σ_v and α_2 are the model's only handles on nonlinear velocities; a velocity effect beyond a $k_\parallel^2$ correction is outside both.

- **Linear response.** The largest absorption-test shifts (1.5–2.6 ellipse units) are linear extrapolations; a refit could move them.

- **The pair list at the σ_v prior edge.** Building the pair list at the prior edge adds the pairs of bins two apart (Section 6); it needs new spectra, the four covariances and new derivative caches, and is queued as an overnight study.

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